



St Catherine's
School
Waverley, Sydney

Student Number: _____

Year 12
Assessment Task 1
27/2/2007

Mathematics Extension II

Student Number _____

Time allowed: 55
minutes

Reading time: NIL

Course weighting:
15%

General Instructions

- Attempt ALL questions
- Write your Student NUMBER at the top of this page and on the writing paper used

Sections

Marks

Total marks

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1, \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Q.1 Find the Argument and modulus of $\frac{(1-i)^4}{(1+\sqrt{3}i)^2}$ (5m)

Q.2

(a) Sketch the locus of z :

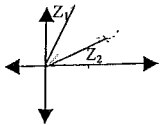
(i) $\arg(z-1-i) = \frac{\pi}{4}$ (2m)

(ii) $\arg(z-2i) = \arg(z+1)$ (2m)

(b) (i) Sketch the locus of z , such that $|z+i| = |z-1|$ (2m)

(ii) Describe the locus and find its Cartesian equation (2m)

Q.3. z_1 and z_2 are two complex numbers such that $|z_1| = |z_2|$, explain why $\frac{z_1+z_2}{z_1-z_2}$ is purely imaginary. (4m)



Q.4. (i) Solve for z , $z^6 = 1$ in the field of Complex Numbers. (2m)

(ii) Factorise $z^6 - 1$ in the field of Complex Numbers. (1m)

(iii) Factorise $z^6 - 1$ in the field of Real Numbers. (2m)

(iv) Explain why the roots of $z^4 + z^2 + 1 = 0$ are among the roots of $z^6 - 1 = 0$ (2m)

(v) State the roots of $z^4 + z^2 + 1 = 0$ (1m)

Q.5. If $z = \cos \theta + i \sin \theta$,

(i) show that $z^n + \frac{1}{z^n} = 2 \cos n\theta$ (2m)

(ii) Hence show that $\cos^4 \theta = \frac{1}{8}(\cos 4\theta + 4 \cos 2\theta + 3)$ (3m)

(note: $(x+y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$)

Q.6. The equation $x^3 + 3x^2 - 2x + 5 = 0$ has roots α, β and γ .

Find the equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$ and $\frac{1}{\gamma}$ (2m)

Q.7.

(i) Given that α is a zero of multiplicity n for a polynomial $P(x)$, Show that α is a zero of multiplicity $(n-1)$ for $P'(x)$ (2m)

(ii) Given that $P(x): 2x^4 - 3x^3 - 3x^2 + 7x - 3$ has a zero of multiplicity 3, factorise $P(x)$. (4m)

Q.8

(i) Show that $\frac{x^3 - 4x - 10}{x^2 - x - 6} = x + 1 + \frac{3x - 4}{x^2 - x - 6}$ (1m)

(ii) Hence express $\frac{x^3 - 4x - 10}{x^2 - x - 6}$ as a sum of partial fractions. (3m)

End of Paper

Q.1

$$|1-i| = \sqrt{2} \quad |1+\sqrt{3}i| = 2$$

$$\text{Arg}(1-i) = -\frac{\pi}{4} \quad \text{Arg}(1+\sqrt{3}i) = \frac{\pi}{3}$$

$$\frac{(1-i)^4}{(1+\sqrt{3}i)^2} = \frac{(\sqrt{2})^4}{2^2} = 1$$

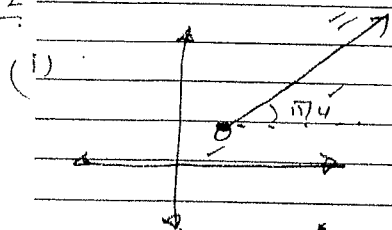
$$\text{Arg} \frac{(1-i)^4}{(1+\sqrt{3}i)^2} = 4 \text{Arg}(1-i) - 2 \text{Arg}(1+\sqrt{3}i) \pm 2n\pi$$

$$= 4\left(-\frac{\pi}{4}\right) - 2\left(\frac{\pi}{3}\right) \pm 2n\pi$$

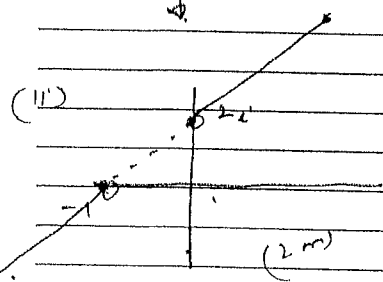
$$= -\pi - \frac{2\pi}{3} \quad (+2\pi)$$

$$= \frac{\pi}{3}$$

Q.2



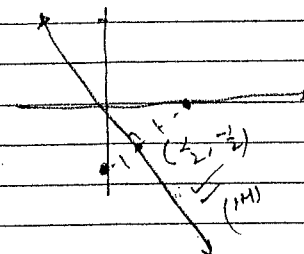
$$\text{Arg}(z - (1+i)) = \frac{\pi}{4}$$



$$\text{Arg}(z - 2i) = \text{Arg}(z+i)$$

Q.3) $|z+i| = |z-1|$

equidistant from $(0, -1)$ and $(1, 0)$
Locus is the perpendicular bisector
of the interval $(0, -1)$ and $(1, 0)$



Cartesian equation is

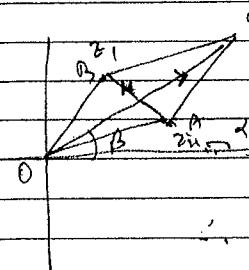
$$y + \frac{1}{2} = -1\left(x - \frac{1}{2}\right)$$

$$2y + 1 = -2x + 1 \quad (1)$$

$$y = -x$$

IM = midpoint
and bisector.

Q.3



note
 $\vec{OC} = z_1 + z_2$
 $\vec{AB} = z_1 - z_2$

$\triangle OAC$ is a rhombus

$\therefore OC \perp AB$ — (1)

$$\text{also Arg} \frac{z_1 + z_2}{z_1 - z_2}$$

$$= \text{Arg}(z_1 + z_2) - \text{Arg}(z_1 - z_2)$$

$$= \beta - \alpha \quad (\text{Ref: fig})$$

$$= -(\alpha - \beta)$$

$$= -90^\circ \quad \text{from (1)}$$

$\therefore \frac{z_1 + z_2}{z_1 - z_2}$ is purely imaginary

Q.4

$$z^6 = 1$$

Let $z = r \text{cis } \theta$

$$|z|^6 = 1$$

$$|z|^6 = 1$$

$$r^6 = 1$$

$$r = 1 \quad (r \text{ is real}) \quad \checkmark$$

$$\therefore z = \text{cis } \theta$$

$$\text{cis } 6\theta = 1$$

$$\cos 6\theta = 1, \quad \sin 6\theta = 0$$

$$6\theta = 0, +2\pi, +4\pi, 6\pi$$

$$\theta = 0, +\frac{\pi}{3}, +\frac{2\pi}{3}, \pi$$

$$\text{cis } 0, \text{cis } \pi/3, \text{cis } (-\pi/3), \text{cis } (2\pi/3), \text{cis } (-2\pi/3)$$

and $\text{cis } \frac{\pi}{n}$ are the roots.

(11)

$$z^6 - 1 = (z-1)(z - \text{cis } \pi/3)(z - \text{cis } (-\pi/3))$$

$$(z - \text{cis } 2\pi/3)(z - \text{cis } (-2\pi/3))(z+1)$$

$$z^6 - 1 = (z-1)(z+1) \left(z^2 - z(\text{cis } \pi/3 + \text{cis } (-\pi/3)) + \text{cis } \pi/3 \cdot \text{cis } (-\pi/3) \right)$$

$$\left(z^2 - z(\text{cis } 2\pi/3 + \text{cis } (-2\pi/3)) + \text{cis } 2\pi/3 \cdot \text{cis } (-2\pi/3) \right)$$

$$= (z-1)(z+1)(z^2 - 2z \cos \pi/3 + 1)(z^2 - 2z \cos 2\pi/3 + 1)$$

$$= (z-1)(z+1)(z^2 - z + 1)(z^2 + z + 1)$$

$$(iv) \quad z^6 - 1 = (z^2)^3 - 1$$

$$= (z^2 - 1)(z^4 + z^2 + 1) \quad \checkmark$$

The roots of $z^4 + z^2 + 1 = 0$ are among the roots of $z^6 - 1 = 0$ $\checkmark$

$$(v) \quad z^4 + z^2 + 1 = 0$$

Roots are

$$\text{cis } \frac{\pi}{3}, \text{cis } (-\pi/3), \text{cis } 2\pi/3, \text{cis } (-2\pi/3) \quad \checkmark$$

Q.5

$$z = \cos \theta + i \sin \theta$$

$$\frac{1}{z} = \cos \theta - i \sin \theta$$

$$z^n = (\cos \theta + i \sin \theta)^n$$

$$= \cos n\theta + i \sin n\theta \quad (\text{De Moivre's}) \quad \checkmark$$

$$\left(\frac{1}{z}\right)^n = (\cos \theta + i \sin \theta)^{-n} \quad \checkmark$$

$$= \cos(n-\theta) + i \sin(n-\theta)$$

$$= \cos n\theta - i \sin n\theta \quad \checkmark$$

$$\therefore z^n + \frac{1}{z^n} = 2 \cos n\theta \quad \checkmark$$

(11) Consider

$$\left(z + \frac{1}{z}\right)^4 = z^4 + 4z^3 \cdot \frac{1}{z} + 6z^2 \cdot \frac{1}{z^2} + 4z \cdot \frac{1}{z^3} + \frac{1}{z^4}$$

$$= \left(\frac{z^4 + 1}{z^4} \right) + 4 \left(\frac{z^2 + 1}{z^2} \right) + 6$$

$$= 2 \cos 4\theta + 4(2 \cos 2\theta) + 6$$

$$\text{also } z + \frac{1}{z} = 2 \cos \theta$$

$$\therefore 16 \cos^4 \theta = 2 \cos 4\theta + 8 \cos 2\theta + 6$$

$$\therefore \cos^4 \theta = \frac{1}{8} (\cos 4\theta + 4 \cos 2\theta + 3)$$

Q. 6

$$\text{Let } P(x) : x^3 + 3x^2 - 2x + 5$$

α, β, γ are roots

$$\therefore P(\alpha) = P(\beta) = P(\gamma) = 0$$

Consider $P\left(\frac{1}{x}\right)$

$$P\left(\frac{1}{x}\right) = P(x) = 0$$

Similarly $\frac{1}{\beta}, \frac{1}{\gamma}$ are roots of $P\left(\frac{1}{x}\right) = 0$

$\therefore \frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ are roots of

$$P\left(\frac{1}{x}\right) = 0$$

$$\frac{1}{x^3} + \frac{3}{x^2} - \frac{2}{x} + 5 = 0$$

$$1 + 3x - 2x^2 + 5x^3 = 0$$

Q. 7

$$P(x) = (x - \alpha)^n \cdot Q(x)$$

Consider

$$P'(x) = (x - \alpha)^n \cdot Q'(x) + Q(x) \cdot n(x - \alpha)^{n-1}$$

$$= (x - \alpha)^{n-1} \left((x - \alpha) \cdot Q'(x) + n Q(x) \right)$$

$(x - \alpha) Q'(x) + n Q(x)$ is a polynomial

$\therefore \alpha$ is a root of multiplicity $(n-1)$ for $P'(x)$

(ii) Let α be the root of multiplicity 3 for $P(x)$

$\therefore \alpha$ is a root of multiplicity 2 for $P'(x)$ and 1 for $P''(x)$

$$P(x) = 2x^4 - 3x^3 - 3x^2 + 7x - 5$$

$$P'(x) = 8x^3 - 9x^2 - 6x + 7$$

$$P''(x) = 24x^2 - 18x - 6$$

$$P''(x) = 0 \Rightarrow 6(4x^2 - 3x - 1) = 0$$

$$6(4x+1)(x-1) = 0$$

$$x = -\frac{1}{4} \text{ or } x = 1$$

$$\text{now } P'(1) = 8 - 9 - 6 + 7 = 0$$

$$\therefore P''(1) = 24 - 18 - 6 = 0$$

$\therefore 1$ is a root of multiplicity 3 for $P(x) = 0$

$$\therefore P(x) = (x-1)^3 \cdot Q(x)$$

$$= (x-1)^3 (2x+3)$$

(by observation)

Q. 8

$$\begin{array}{r} x+1 \\ x^2-x-6 \overline{) x^3-4x-10} \\ \underline{x^3-x^2-6x} \\ x^2+2x-10 \\ \underline{x^2-x-6} \\ 3x-4 \end{array}$$

$$\therefore \frac{x^3-4x-10}{x^2-x-6} = x+1 + \frac{3x-4}{x^2-x-6}$$

Consider

$$\frac{3x-4}{(x-3)(x+2)} \equiv \frac{A}{x-3} + \frac{B}{x+2}$$

$$3x-4 \equiv A(x+2) + B(x-3)$$

If $x=3$, $5 = 5A \quad \therefore \underline{A=1}$

If $x=-2$, $-10 = -5B \quad \therefore \underline{B=2}$

$$\text{Thus } \frac{x^3-4x-10}{x^2-x-6} = (x+1) + \frac{1}{x-3} + \frac{2}{x+2}$$